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A new way of implementing prices - a Dynamic Pricing Strategy in E-commerce

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Abstract

Dynamic Pricing is an emerging subject that has been deeply explored during the Covid19 pandemic. The project Dynamic Pricing Model in E-commerce consists in designing a challenging model to apply to the Navigator Hub (digital platform), as prices are still manually adjusted. “Does customer data analytics from the online channel suggest there is room for customer differentiation in real time?” is, therefore, the key research question. The main focus was on one possible variable for the model, a cluster analysis using IBM SPSS. Four segments, exhibiting different customers’ patterns, arise from there, thus, confirming there is margin to adjust prices.

Keywords: E-commerce, Digital, Strategy, Revenue, Business to Business, Online, Cluster, Client.

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1. INTRODUCTION

a. Company overview

The Navigator Company is an integrated producer of forest, pulp, paper, tissue, and bioenergy, whose business is leveraged by state-of-the-art factories on a worldwide scale, with cutting-edge and sophisticated technology.

The Company's business model revolves around a raw material of excellence – Eucalyptus globulus – whose intrinsic features allow a strategy of differentiation to be pursued, based on high-quality products. This allows Navigator to be an international benchmark in its sector. Thus, the Company is the largest manufacturer in Europe of uncoated wood free paper (UWF) for printing and writing (the category that includes office paper) and the 6th largest producer worldwide. It is also the leading European manufacturer of bleached eucalyptus pulp (BEKP). From a national perspective, it is the Portugal's third-largest exporter, accounting for approximately 1% of the country's GDP and around 2,5% of all exports of national goods. Navigator markets products under various brands, including Pioneer, Inacopia, Explorer, Inaset, Soporset, Target, Discovery, and Navigator (the global best seller in the premium office paper segment). The company operates several industrial manufacturing facilities, including Setúbal, Figueira da Foz, Aveiro and the Vila Velha de Rodão pulp and paper mill. It also offers production planning and logistics services. Additionally, the company operates in 130 countries across the five continents, and it has its headquarters located in Setúbal, Portugal.

Navigator currently has six business units, including Research & Development, Agro-Forestry,

Pulp production & sales, Paper production & sales, Tissue production & sales and Energy.

b. Industry overview

The paper industry has been undergoing a huge transformation, mainly during the last years.

Major changes in physical stores and e-commerce, as well as innovative prospects for new growth have been made across the paper and forest-products industry, in order to face the lower global demand. However, according to *Appendix 1*, the global paper and paperboard industry continues to grow, despite a decline in the graphic-paper segment.

Companies in this sector have been analyzing new innovative ways to boost their profitability.

Packaging, tissue papers, pulp for hygiene products and for textile applications are the more recent trends that involve sustainability matters and, thus, have been growing around the world.

Following some research, graphic papers continue to face a worldwide decline in demand by around 7,6%, which tends to be accelerating and, thus, it requires innovation and a restructure of the production capacity. Consumer packaging and tissue, as well as transport and industrial packaging are value-creating sectors with opportunities to grow even more, through demographic shifts and consumer trends. Also, fiber has been facing turbulent times and some volatility in prices, spreading across different locations.

Therefore, connecting sustainability requirements, e-commerce, and technology integration require facing future challenges. The board of companies should be concerned about finding the next level of cost optimization and value-creating growth roles for forest products. In this constantly changing business, increasing relevancy in the packaging world, and finding the right

path in next-generation bio-products are crucial matters to develop this industry that seemed to be disappearing (Berg and Lingqvist 2019). It is also fundamental to ensure the right offer is available to the right customer at the right time, and this implies examining customer data on the one hand, and dynamic pricing models, on the other hand.

The study aim of this project is therefore to explore - and materialize, where possible - the opportunities for applying dynamic business pricing in Navigator's online channel. The next section, Literature Review, will then include a detailed knowledge and researched about the main benefits, drawbacks, and limitations of this pricing strategy as well as, pricing decisions in Business-to-Business context and commodities industries. Furthermore, some solutions/different models and the implementation process (including potential variables) will be discussed. Afterwards, in methodology, through the historical sales data of the Company, a clustering analysis in IBM SPSS will be done in order to find the best clients' segments for the company, that will lately constitute a variable for the whole model, indicating in which of them prices could be set slightly above or below. These segments are going to be discussed in "Analysis and Discussion" section, while presenting valid reasons and rationales for their similar behaviors. Lastly, some recommendations on other potential global variables to be included in the model, as well as main conclusions will be gathered to present to the Company.

2. LITERATURE REVIEW

a. Pricing decisions in emerging markets

With the increasing complexity of the markets, pricing decisions are not static and more

challenging, therefore, there is the need to continuously analyze the market and the competition. As a consequence, many companies are adopting computerized pricing support systems (Phillips 2021, Chapter 2.4.3). Dynamic pricing consists of the utilization of Artificial Intelligence to establish different prices for the same product to different customers, depending on exclusive personal and global factors (Enache 2021, 1). This high-tech and fast spreading pricing strategy is mainly utilized in the e-commerce sector, where companies can collect market data and adjust the products' prices to the customers' willingness to pay. In this sector, according to *Appendix 2*, more than 35% of companies in North America and Europe either use this model or plan to use it, soon (Dearth 2022).

b. Dynamic Pricing Model

The concept of dynamic pricing was introduced during the 1980s by the airline industry. After that, in the 1990s, the internet and new technologies emerged, incentivizing hotels to implement this price strategy as well. Later, by 2000, e-commerce was introduced, making huge improvements in the dynamic pricing model, by Amazon and eBay (Baldwin 2019). Nowadays, dynamic pricing is a well-known and successful topic that is very demanding in digital platforms, across many different industries (for instance, in the retail environment). It has emerged exponentially since the Covid19 pandemic, not only in online channels, but also in offline ones. Researchers have been studying and analyzing strategies and approaches to successfully implement this model in the e-commerce era.

c. Main goals and future achievements

There are goals that enterprises want to achieve when using dynamic pricing models, that are mainly to predict the prices that each customer would pay for a product and to have a fast response to market fluctuations. Consequently, the company's profitability will increase, as well as customer engagement and loyalty (Enache 2021, 1).

This pricing module has been proven efficient in increasing levels of customer satisfaction and, consequently, promoting sales growth (at least 2%-5%) and sales margins improvement (at least 5%-10%), as illustrated in *Appendix 3* (BenMark, Klapdor, Kullmann, and Sundararajan 2017).

d. Benefits and Drawbacks

As successfully proved, there are many advantages of this type of pricing model, and they are related to both the company's performance indicators and customer satisfaction. This model allows the company to make quick price reductions and to adjust its prices based on an individual customer's likelihood of purchasing a product, instead of the market (Biscontini 2020). Also, the use of Artificial Intelligence to establish prices (through elastic or competitive models) allows the company to get market information, that can be used to analyze consumer behavior and make better marketing decisions, rather than pricing (Enache 2021, 3).

However, negative aspects may also arise. There is always some level of uncertainty about customers' responses to different prices (Enache 2021, 3). In addition, this strategy may lead to unfair discrimination between consumers and boost collusion between firms, which can then lead to an increase in prices (Thornhill 2022).

e. Challenging the Business-to-Business context

According to Fatemeh Goli and Manijeh Haghighinasab, the dynamic price has received relatively little attention in the Business-to-Business environment. It was an under researched topic that, however, has been growing during this new digital era (Goli and Haghighinasab, 2021, 14).

B2B situations are particularly different from B2C ones. In the first case, sellers can easily vary prices, not only across buyers, but also between subsequent purchases of the same buyer. In addition, trust and commitment between sellers and buyers are crucial to determine long-term and stable relationships between them. That is why prices also have long-term effects in B2B markets. In general, transactions can be more complex and exceptional situations, such as a pandemic, can influence buyers' decisions. Overall, in this business environment, decision makers (meaning, buyers and sellers) are assumed to behave rationally, which might be different in B2C cases (Zhang, Netzer and Ansari 2014, 1).

Therefore, making pricing decisions in B2B situations can be very challenging. Firstly, pricing must consider "unobserved random shocks", such as demand boosts or supply shortages. Secondly, maintaining confidence and commitment across all decision makers' relationships can be difficult due to environmental uncertainty. Moreover, segmenting and targeting B2B markets evolves a lot of variables' analyses. Segmenting depends on customer size, industry, customer behavior and buying behavior, whereas targeting carefully analyses the price sensitivity of each buyer, splitting them between "programed" and "transactional" buyer (Zhang, Netzer and Ansari 2014, 3). Thus, changing price decisions in a B2B operational

company requires adaptability, market research and alignment between all the departments. Companies should focus on who is responsible for what, how to make appropriate discounts (right parameters), as well as aligning them across the organization (Phillips 2021, Chapter 2.1).

f. Commodity Markets

Regarding the commodity industry, there are some multinational companies that are already using dynamic pricing strategies, for instance, Walmart and Amazon, or even gas stations (Morenne 2022). However, it is still an underexplored topic due to the high volatility in commodity prices. Companies in this business must spend a large amount on fixed assets to turn raw materials into finished products. Therefore, their products compete on price and differentiation (Bank 2022).

Global macroeconomic shocks, such as the pandemic recession, faced demand shrinkages and supply disruptions which, in turn, lead to a collapse in the commodity prices, as it can be seen in *Appendices 4 and 5* (Vasishtha 2022).

Consequently, a Commodities Factor Model (CFM) has been improved as a dynamic factor model, for a large cross-section of energy and non-energy commodity prices. The model is decomposed according to *Appendix 6*, and it relies on available market data, making it easier to maintain and adjust prices and/or groupings. Thus, it imposes a block programming structure on the factors that decompose the price movement of each commodity into a global, block or idiosyncratic component. The global factor captures price trends that are common to all included commodities, whereas the idiosyncratic captures supply shocks and block factors